Mengmeng Wang

Graduate Researcher, Data Scientist

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Profile

I am a dedicated statistical signal processing researcher & data specialist with a background in engineering and computer science. I enjoy solving challenging problems in statistical data modelling, big data analytics and machine learning. I have developed ML algorithms, time-series analysis, predictive modelling, raw data transformation & analysis, and image processing algorithms. I am experienced in Python, MATLAB, Excel, R, Pandas, NumPy, Scikit-Learn, NLTK, SciPy, Matplotlib, JSON, RegEx, Keras, Power BI, and Tableau.

Education

Doctor of Philosophy – Biomedical Engineering 2017 – Present
The University of Melbourne

Master of Science – Signal Processing 2013 – 2014
University of Bristol

Bachelor of Engineering – Telecommunications 2009 – 2013
Beijing University of Posts and Telecommunications

Experience

Graduate Researcher April 2017 – Present
The University of Melbourne

- PhD Project: signal analysis techniques in resting state functional near-infrared spectroscopy.
- Applied quantitative mathematical modellings to model functional brain activities.
- Investigated the impact of noise on brain signals and analysed brain functional connectivity using signal processing and statistical tools.
- Developed 3+ statistical algorithms from pure mathematical derivations to MATLAB implementations.

Data Processing & Machine Learning Tutor (Casual) March 2019 – Present
The University of Melbourne

- Tutored 3 university-level subjects: machine learning, elements of data processing, signals and systems.
- Taught data wrangling, format, visualisation, natural language processing, supervised and unsupervised learning, neural network.
- Used Python and Pandas, NumPy, Scikit-Learn libraries to teach advanced machine learning and modern data science concepts.

Consulting Engineer May 2015 – April 2017
China Academy of Information and Communications Technology

- Analysed domestic and international technologies in telecommunications and information science.
- Provided strategic advice on markets and created technical models and evaluation tools.
- Produced over 20 consulting reports and data analysis deliverables to governmental and industrial clients.

Selected Projects (Available on GitHub)

Financial Timeseries Processing and Forecast

- Performed financial data analysis in Level 1 Limit Order Book data, across data processing pipeline.
- Data cleaning and preparation: outlier detection, data visualisation and feature engineering.
- Data analysis: correlation, moving-average, auto-regression analysis.

- Timeseries forecasting: auto-regression model and machine learning models (decision tree, logistic regression, neural networks).
- Tools used: Python, Jupyter Notebook, Pandas, NumPy, matplotlib, seaborn, statsmodels, sklearn, keras.

Customer Purchasing Behaviours Analysis

- Analysed customer transaction and purchase behaviour data to find patterns in customer behaviours.
- Perform data cleaning, preparation and visualisation to facilitate analysis.
- Draw insights on change in store layout in customer behaviours.
- Tools used: Python, Jupyter Notebook, Pandas, NumPy.

Reinforcement Learning and Multi-Armed Bandits (MABs)

- Implementation and evaluation of state-of-art reinforcement learning algorithms: ϵ-Greedy, Upper Confidence Bound (UCB), LinUCB and KernalUCB.
- Tools used: Python, Jupyter Notebook, NumPy.

Publications

- **M. Wang**, C. Davey and L. Johnston, "Correction of induced functional connectivity in filtered resting state fNIRS data," *The 27th Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, 2021.
- **M. Wang**, L. Johnston and C. Davey, "Correction for time-varying signal power in fNIRS connectivity analyses," *Society of fNIRS Virtual Conference*, 2021.
- **M. Wang** and A. Seghouane, "Motion Artefact Removal in Functional Near-infrared Spectroscopy Signals Based on Robust Estimation," *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2019.
- **M. Wang**, F. Zhang and D. Agrafiotis, "A very low complexity reduced reference video quality metric based on spatio-temporal information selection," *IEEE International Conference on Image Processing*, 2015.

Certificates and Awards

- Online Course Certificates in Data Science 101, Data Science Tools, Big Data 101, Introduction to Cloud, Blockchain Essentials
- HackerRank Programming Certificates for Python and SQL
- Virtual Internship Certificates from ANZ Australia, Goldman Sachs, JPMorgan Chase & Co.
- Virtual Internship Certificates from KPMG Australia, Deloitte Australia, Quantium, General Electric
- Conference Travel Scholarships, The University of Melbourne
- Melbourne Research Scholarships, The University of Melbourne
- Excellent Examination Performance Prize, University of Bristol

Skills

Programming Languages Python, Jupyter Notebook, SQL, MATLAB

Technical Tools GitHub, Tableau, Power BI, Excel

Python Libraries Pandas, NumPy, Scikit-Learn, NLTK, SciPy, Matplotlib, JSON, RegEx

Professional Data Analysis, Technical Writing, Project Management, Teamwork, Communications

Volunteer Experience

Girl Power Mentor March 2021 – Present

The University of Melbourne

- Mentored Year 11/12 female students with interests in science and engineering.
- Coached girls to develop strength and confidence to pursue tertiary education in science and engineering.

References

Available Upon Request